Kaarigar Setu: folder-by-folder breakdown

What each folder of the project holds and what it does. Paths are relative to `Karigar-Setu-rebuilt/`.

1. `kaarigarsetu-java/` (the web application)

1.1 `src/main/java/com/kaarigarsetu/` (144 files in 17 packages)

Package	Files	What it does
`model/` (18)	`Worker`, `Employer`, `Assessment`, `AssessmentAnswer`, `Attestation`, `AttestationDetail`, `WorkerConfirmation`, `FlaggedAttestation`, `FraudFlag`, `TrustEdge`, `TrustEdgeRow`, `Credential`, `CredentialRecord`, `DisputeCase`, `Moderator`, `ModeratorAction`, `TrustedDevice`, `AuditLog`	Java records: the plain data shapes
`exception/` (15)	`InvalidOTPException`, `DuplicateAttestationException`, `LowTrustScoreException`, `ConsentRequiredException`, `UnintelligibleAudioException`, `ExternalServiceException`, `ConflictException`, `NotFoundException`, `ValidationException`, `ForbiddenException`, `RateLimitExceededException`, `AuthenticationFailedException`, `ChallengeException`, `ProofOfWorkRequiredException`, `DataAccessException`	our own error types
`config/` (5)	`AppSettings`, `DatabaseConfig`, `PooledConnectionProvider`, `ConnectionProvider`, `Transactions`	environment settings, connection pool, transactions
`util/` (5)	`Json`, `CanonicalJson`, `WavValidator`, `AuditLogWriter`, `CredentialExporter`	JSON, signing format, audio check, file writing
`dao/` (14)	`JdbcDaoSupport` plus `WorkerDao`, `EmployerDao`, `AssessmentDao`, `AttestationDao`, `TrustEdgeDao`, `CredentialDao`, `FraudFlagDao`, `OtpChallengeDao`, `TrustedDeviceDao`, `ModeratorDao`, `ModeratorActionDao`, `AuditLogDao`, `ReplayDao`	all SQL, one class per table
`assessment/` (19)	`AssessmentService` (abstract) and six trade subclasses (`Electrician`, `Plumber`, `Carpenter`, `Mason`, `Welder`, `Painter`), `AssessmentScoringDispatcher`, `AssessmentPipelineService`, `AssessmentServlet`, `AssessmentResult`, `Band`, `SkillScoring`, `SkillProfile`, `TradeCatalog`, and the replay-check classes `ReplayDetector`, `ReplayChallenge`, `SpokenCode`, `TextShingles`	the voice test and its scoring
`worker/` (2), `employer/` (2)	`WorkerService`, `WorkerServlet`, `EmployerService`, `EmployerServlet`	registration and the passport

Package	Files	What it does
`attestation/` (8)	`AttestationWorkflowService`, `AttestationServlet`, `ConfirmationResponseService`, `AttestationRules`, `AttestationService`, `AttestationStore`, `AttestationRepository`, `DaoAttestationRepository`	employer confirmation; the worker's accept or dispute
`trust/` (7)	`ReputationScoring`, `TrustScoreService`, `TrustGraphService`, `TrustBand`, `Edge`, `TrustGraphRepository`, `DaoTrustGraphRepository`	the trust formula and the endorsement graph
`credential/` (3)	`CredentialService`, `CredentialServlet`, `QRGenerator`	issuing, checking and the QR code of the signed credential
`moderator/` (3)	`ModeratorService`, `ModeratorServlet`, `SecurityMonitor`	staff queues, two-person approval, the overview
`web/` (10)	`AppContext`, `AppBootstrap`, `BaseServlet`, `ApiErrors`, `AuthServlet`, `PageServlets`, `Sessions`, `CsrfFilter`, `SecurityHeadersFilter`, `SecurityEvents`	request entry and exit, application wiring, sessions
`security/` (24)	`AuthFilter`, `AuthService`, `OtpService`, `OtpSender`, `ConsoleOtpSender`, `HttpSmsOtpSender`, `OtpChallengeStore`, `InMemoryOtpChallengeStore`, `PasswordHasher`, `PhoneProtector`, `ProofOfWork`, `RateLimiter`, `Role`, `AuthenticatedUser`, `SignatureService`, `PresentationTokens`, `DataCipher`, `DataKeys`, `SealedDataException`, `AuditLogger`, `AuditLogRepository`, `DaoAuditLogRepository`, `AuditFileSink`, `ChainedAuditFileSink`	sign-in codes, access rules, signing, encryption, audit
`ml/` (2)	`MlGateway`, `MlServiceClient`	the link to the Python service
`privacy/` (1)	`ErasureService`	deleting a person's data
`admin/` (5)	`ModeratorAdminCli`, `DemoSeed`, `AuditVerifyCli`, `SealAtRestCli`, `SealedColumns`	operator tools
root	`HealthServlet`	the `/health` check

1.2 `src/main/webapp/` (what the user sees)

Folder	Files	What it does
`WEB-INF/jsp/`	`index`, `worker-dashboard`, `assessment`, `employer-dashboard`, `verify`, `demo`, `error`	the 7 pages
`static/js/`	`common`, `auth`, `pow`, `worker`, `assessment`, `wav`, `employer`, `demo`	the page scripts: shared helpers, sign-in, the passport, the voice test, audio conversion, the employer form

Folder	Files	What it does
`static/css/`	`app.css`	the single stylesheet
`WEB-INF/`	`web.xml`	the filters, JSP rules and error pages

1.3 Other parts of this folder

Path	What it holds
`src/main/resources/trades.json`	the six trades, in English and Hindi
`src/test/java/`	64 test files, 498 tests, grouped by `assessment`, `attestation`, `trust`, `credential`, `moderator`, `security`, `dao`, `web`, `util`, `architecture`, `privacy`, `admin`, `config`
`pom.xml`, `pom-rb.xml`	the build files (`pom-rb.xml` is the development run)
`spotbugs-exclude.xml`	static-analysis exceptions, each with its reason
`Dockerfile`	builds the application image

2. `kaarigarsetu-ml-service/` (the Python service)

Path	What it holds
`app/main.py`, `config.py`, `security.py`	the FastAPI app, settings, the shared-secret check
`app/routes/`	`transcribe`, `audio` (fingerprint), `evaluate`, `rubric`, `fraud`, `trust`
`app/services/`	speech to text: `faster_whisper_client`, `whisper_client`, `stub_transcriber` · scoring: `evaluation_service`, `rubric_evaluator`, `rubric_store`, `guard`, `llm_client` · `audio_fingerprint` · `fraud_detector` · `graph_engine` (PageRank)
`app/models/`	six rubric files plus `_common.json`; the trained fraud model with its checksum and metrics; the language-model prompts
`app/training/`	scripts that train and check the fraud model on synthetic data
`tests/`	398 tests
`benchmarks/`	speech-accuracy and replay-check benchmarks

3. `kaarigarsetu-moderator-console/` (the JavaFX staff tool)

12 files in `src/main/java/com/kaarigarsetu/console/`, 18 tests.

Files	What they do
`ModeratorConsoleApp`	the window: sign-in and the five tabs (Fraud review, Employers, Disputes, Approvals, Security)
`Launcher`	starts the application
`ModeratorApi`, `HttpModeratorApi`, `ModeratorIdentity`	the interface to the Java API, its HTTP implementation, who is signed in
`FlaggedRow`, `EmployerRow`, `DisputeRow`, `PendingRow`	one row of each table
`SecurityOverview`, `ActionResult`, `ConsoleException`	the overview data, action results, console errors

4. `database/`

File	What it holds
`schema.sql`	the first tables
`migrations/002` to `011`	the pipeline columns; skill bands; employer ratings; employer vetting; worker disputes; the counted-confirmations view; named moderators and two-person approval; trusted devices; encrypted columns; replay-detection tables
`setup_roles.sql`	the restricted database role the application uses

5. `scripts/`

Group	Files
Set-up and run	`dev-setup.sh`, `dev.sh`, `dev-https.sh`, `dev_https_proxy.py`, `console.sh`, `compose-setup.sh`, `demo-seed.sh`, `otp.sh`
Tests	`e2e.py` (192 checks), `api_matrix.py` (92), `security_attack_demo.py` (57 attacks), `coverage.sh`, `pow_js_check.js`
Credentials and keys	`verify_credential.py`, `QrDecode.java`, `generate-signing-key.sh`, `rotate-signing-key.sh`
Operator tools	`moderator-admin.sh`, `seal-at-rest.sh`, `audit_decrypt.py`
Other	`demo_fraud_burst.py`, `llm_check.py`, `osv_scan_maven.py`

6. Deployment

Path	What it holds
`docker-compose.yml`	four services: database, ML service, application, HTTPS proxy

Path	What it holds
`docker/postgres/init.sh`	creates the schema, applies the migrations, creates the restricted role
`docker/tomcat/server.xml`	hardened Tomcat settings
`docker/app/entrypoint.sh`	starts the application container
`docker/caddy/Caddyfile`	the HTTPS proxy
`.github/workflows/ci.yml`	the build and test pipeline
`.env.example`	the settings template

---

7. `docs/`

Group	Files
Design	`PROJECT_OVERVIEW`, `TRUST_SCORE`, `TRUST_AND_NETWORK_SCORES`, `CREDENTIAL_TRUST`, `REPLAY_DETECTION`
Reference	`API_REFERENCE`, `SECURITY_RUNBOOK`, `PRIVACY_DPDP`
Security	`SECURITY_ARCHITECTURE`, `STRIDE_THREAT_MODEL`, `PENTEST_REPORT`, `PENTEST_RESULTS`, `SECURITY_SCAN_REPORT`
Testing and status	`TEST_REPORT`, `IMPLEMENTATION_STATUS`
Review material	`IMPLEMENTATION_WALKTHROUGH` (md and PDF), `PROJECT_REVIEW_ROLES` (md and PDF), `CONTRIBUTIONS`, `FOLDER_BREAKDOWN`, `review-screenshots/`, `screenshots/`

---

8. Root

File	What it is
`README.md`, `README_JAVA_FEATURES.md`	the overview and the Java-features guide
`.env.local`, `data/`	development secrets and runtime data (not meant to be shared)